

Comparing Context-sharing Interfaces in XR Remote Collaboration

Eunhee Chang^{*†}
University of South Australia

Yongjae Lee^{*‡}
Arizona State University

Byoungyun Yoo[§]
Korea Institute of
Science and Technology

Hojeong Im[¶]
Korea Institute of
Science and Technology

ABSTRACT

eXtended Reality (XR) remote collaboration refers to working together on virtual data or real objects through various types of reality. To achieve a higher immersive collaboration experience, it is critical to converge heterogeneous realities in one shared workspace, where environmental changes are continuously updated. In this study, We extend our previous work, the Webized XR system, in terms of sharing the local environmental context. This system can provide three types of context-sharing interface (2D video, 360 video, and 3D reconstruction), delivering spatial information about the workspace. We also performed a pilot user study to measure the quality of this system.

Index Terms: Human-centered computing—Virtual Reality; Human-centered computing—Mixed augmented reality; Human-centered computing—User studies;

1 INTRODUCTION

Remote collaboration is a reciprocal action that two or more far-off workers cooperate to achieve their common goal. The application of eXtended Reality (XR) technology to remote collaboration, which embraces Virtual Reality (VR) and Augmented Reality (AR), has been actively studied since immersive experience through the XR can reduce unnecessary communication between workers and improve work efficiency overall.

Though various methods for XR remote collaboration have been proposed, it is hard to exploit the systems in a real situation due to their low capabilities and complicated hardware compositions. For example, the works of Piumsombon et al. [2] or Teo et al. [3] provide the remote expert with a real-time 360-degree video and free switching between the video and the 3D model. However, introducing additional sensors inevitably increases hardware complexity. Moreover, the absence of real-time reflection of the local space to the 3D model restricted the studies [3, 4] to designing simple tasks such as just searching for something or moving flat objects.

We previously suggested a brand-new XR remote collaboration system, the Webized eXtended Reality (WXR) [1]. The WXR recognizes objects of interest and synchronizes their location in AR and in VR. This approach enables remote workers to work together on searching and moving volumetric objects only with a camera.

In this study, we extended the WXR. The improved WXR serves two additional context-sharing interfaces (2D video and 360 video) as well as the basic context-sharing interface (3D reconstruction) so that the remote expert captures the local context. In addition, we conducted a pilot study to measure the quality of remote collaboration depending on the interfaces. Using each context-sharing interface, participants performed an object-searching-and-moving scenario as a remote expert. To the best of our knowledge, our study

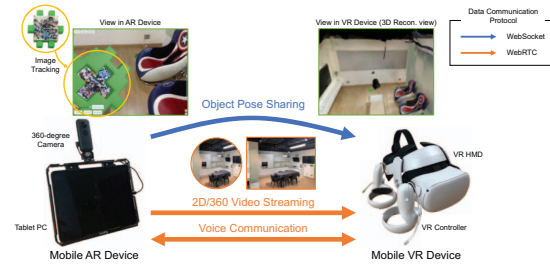


Figure 1: WXR system overview

is the first to measure the XR remote collaboration performance with the scenario of moving volumetric objects, which is practical in the real world.

2 WXR SYSTEM

Fig. 1 shows the overview of the WXR system. A local worker experiences the AR environment using a tablet (iPad Pro Gen. 4) and shares the current poses of objects in the local space with a remote expert by tracking the objects. Alternatively, live videos of a built-in camera and an omnidirectional camera (Insta360 One X) can be transmitted. A remote expert wears VR HMD (Meta Quest 2) and experiences the local space through a 2D video, 360-degree video, and 3D reconstructed virtual scene. The tablet and VR HMD are connected by WebSocket for pose data sharing and by WebRTC for media data (video and audio) streaming. Since our system is developed upon the web standards, it has the advantages of hardware independency and low-latency networking.

2.1 XR Remote Collaboration

In a collaboration with the WXR, a remote expert and a local worker are represented to each other as a virtual avatar, and real objects in the local space are synchronized with their virtual replicas so that they show the same pose. The 3D reconstruction view, which shows the virtual world composed of the replicas, gives a remote expert hints about the arrangement of the local space. Understanding the local context through this view, the remote expert judges the situation of the local space and gives instructions to the local worker. Since the remote worker's virtual world is kept updated with the environmental change by the local worker's action, the remote expert can keep giving correct instructions based on up-to-date information about the local space.

2.2 Context-sharing Interfaces

In addition to the 3D reconstruction view, we added two interfaces: 2D video and 360 video. These features are harmonized with the earlier WXR system [1] rather than ad-hoc so that the remote expert freely switches and experiences the desired context-sharing interface.

2D video: The 2D video context-sharing interface streams the local worker's display, which shows augmented content upon the 2D video frame filming the local space, to the remote expert. The expert watches the local worker's display rendered on a 2D rectangle in a virtual space with a black background.

^{*}denotes equal contribution

[†]e-mail: eunhee.chang@unisa.edu.au

[‡]e-mail: yongjae@asu.edu

[§]e-mail: yoo@byoo.net

[¶]e-mail: hojeong.im@wrl.onl

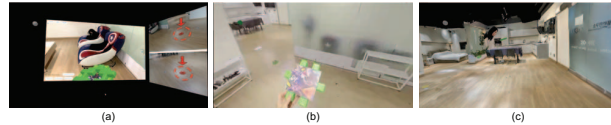


Figure 2: Screen captures of the three context-sharing interfaces: (a) 2D video, (b) 360 video, and (c) 3D reconstruction

360 video: For the 360 video context-sharing interface, the local user transmits the 360-degree video to the remote expert. The 360-degree video is calibrated, so the screen viewed by the remote expert is invariably fixed to the 360-degree camera's orientation. Since the 360 video is rendered egocentrically to the expert, the VR worker can voluntarily check the local situation by changing the view direction.

3D reconstruction: Through the 3D reconstruction context-sharing interface, the local space is represented by a 3D scanned and reconstructed model. Poses of the virtual objects in the remote expert's VR world are kept updated when the local user's AR device tracks the corresponding real ones. The local worker's avatar locates at the same position in the local space and moves along with the local worker.

3 PILOT STUDY

To compare the effectiveness of context-sharing interfaces, we designed a volumetric object searching and navigation task and conducted pilot user study. We recruited 4 participants who did not have any prior spatial knowledge of the local workspace to clarify the effect of context-sharing interfaces on task performance and instructed them to behave as a remote expert in the experiment.

3.1 Design

The user study was conducted in an experimental space called "Living Lab," which simulated a home environment. The Lab is divided into two areas: a residential area (11.1m x 8.4m) and an operational area (2.4m x 1.8m). We intended to set up a spacious and complex area to clarify the effect of context-sharing interfaces on task performance and users' preferences.

We designed the task to entail geometric changes in the local space. To be specific, participants were asked to explore the residential area and find distributed seven waffle blocks, watching the area through the three context-sharing interfaces. To see another side of the area, the participants may ask the local worker to move around (2D or 360) or just move their position in the virtual world by VR controller (3D). Once the participants find a waffle block and the local worker recognizes it by their AR device, a destination for the block appears on the participants' screen. Then, the participants again instructed the local worker to put the block to the destination. The task ends when the local worker moves all the blocks to the designated places. During the experiment, the participants were allowed to ask action to the local worker only with a set of restricted verbal commands (e.g., "forward four steps," "turn forty-five degrees to the right," "recognize the block by the device").

We quantified the quality of remote collaboration by measuring following indices: completion time, total distance of a local worker, system usability scale (SUS), and simulator sickness questionnaire (SSQ). Since the results of the Shapiro-Wilk test indicated violations of the normal distribution, we performed a Friedman test to identify the interface effect on task performance.

3.2 Result & Discussion

According to the Friedman test, we did not find any significant difference between context-sharing interfaces in each measurement. For the completion time, participants spent 460.0 sec (2D), 496.5 sec (360), and 527.0 sec (3D) on average to accomplish task goals

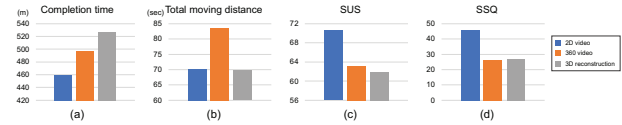


Figure 3: Bar graphs of each measurement: (a) Completion time, (b) Total moving distance, (c) SUS, and (d) SSQ

using each interface. The local worker navigated the residential area for 70.2 m (2D), 83.5 m (360), and 69.7 m (3D), respectively. In terms of system usability, the SUS score indicated 70.6 (2D), 63.1 (360), and 61.9 (3D) for each interface. Lastly, the total SSQ score showed 45.8 (2D), 26.2 (360), and 27.1 (3D), respectively. Fig. 3 illustrates the bar graphs of each index.

According to the interview after experiments, participants reported that they felt difficulty in manipulating VR controllers. Compared to the 2D and 360 video interfaces, the 3D reconstruction interface requires participants to interact more in the VR workspace. Thus, a lack of experience using VR devices might lead to a longer completion time and lower SUS score in the 3D reconstruction condition. Based on this report, we plan to adopt an additional tutorial session before the experiment to help participants to be familiar with manipulating devices. Moreover, a larger sample size is required to clarify the interface effect on task performance.

4 CONCLUSION

Sharing a working context can enrich the immersive experience during XR collaboration. The present study contributes to improving the conventional WXR system by implementing additional context-sharing interfaces. This system provides spatial information about the workspace through 2D video, 360 video, or 3D reconstruction. We modeled the pilot study on the situation (moving volumetric objects) frequently occurring in a collaboration. According to the pilot study, participants showed different responses to using each interface in terms of objective and subjective perspectives. Further study is required to clarify the interface effect on task performance.

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